

A non-even near-ball branch for the L_p –Christoffel–Minkowski equation

Candidate partial result for arXiv:2411.17896

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Status. Candidate partial result, likely valid. The global non-even uniqueness question is *not* claimed. We prove a full non-even local branch, uniqueness already under C^0 closeness to the ball, and automatic symmetry of that branch for even data.

1 Source question

For a support function h on $\mathbb{S}^{n-1}$, write

$$\mathcal{A}h = \nabla_{\mathbb{S}^{n-1}}^2 h + hI$$

and let $s_j(h, \cdot)$ be the normalized j th elementary symmetric function of the eigenvalues of $\mathcal{A}h$. Patsalos and Saroglou study

$$h^{1-p}s_j(h, \cdot) = g, \quad 1 \leq j \leq n-2, \quad 0 \leq p < 1. \quad (1)$$

Their Theorem 1.4 proves small-data uniqueness in the even class and asks whether evenness can be removed [1].

The literal “natural question” about a universal intrinsic-volume L_p –Brunn–Minkowski inequality, appearing earlier in the source, is answered negatively by the source’s own Theorem 1.1. It is not our target.

2 Proof intuition

Evenness is not needed at the linearized level. At the ball, the only new modes introduced by dropping evenness are the degree-one spherical harmonics, corresponding to translations. Their linearized eigenvalue is $1-p$, which is strictly positive. Constants also have positive eigenvalue, while all degrees at least two have negative eigenvalue. Thus the full nonlinear map is locally invertible on all Hölder functions.

There is a second point. A solution assumed only C^0 -close to the ball is automatically Hölder-close, because support functions have controlled Lipschitz norm. Its ordinary area density $s_j(h) = gh^{p-1}$ is then close to 1. The classical $p=1$ Christoffel–Minkowski problem recovers h with two derivatives, modulo translation; the C^0 hypothesis controls that translation. This places the solution inside the full inverse-function neighborhood.

A close inspection of the proof by Ivaki and Milman [53] shows that a slightly stronger statement is actually true. We state it in Theorem D (see next section).

Our third main result can be viewed as a stability result of Theorem C.

Theorem 1.4. *Let $\alpha \in (0, 1)$ and $p \in [0, 1)$, $j \in \{1, \dots, n-2\}$, such that $(p, j) \neq (0, 1)$. Then, there exists $\delta = \delta(n, p) > 0$, such that if g is an even function defined on $\mathbb{S}^{n-1}$, then equation (9) has a unique even classical solution, provided $\|g - 1\|_{C^\alpha} < \delta$.*

We are not aware of any results that indicate whether Theorem 1.4 holds or not without the restriction $\|g - 1\|_{C^\alpha} < \delta$.

Results in the style of Theorem 1.4 have previously been obtained for the L_p -Minkowski (case $j = n-1$) problem. See Chen, Huang, Li, Liu [21] for the symmetric case and Chen, Feng, Liu [20], Böröczky, Saroglou [12] for the extension to the non-symmetric case. The proof in the case $j < n-1$ requires some new ideas and is more elaborate than the corresponding (symmetric) case $j = n-1$. Nevertheless, the question of whether Theorem 1.4 holds without the evenness assumption is, to the best of our knowledge, open and (in our opinion) deserves some attention. Some of the main ingredients for the proof of Theorem 1.4 are Theorem D below by Ivaki and Milman and a non-degeneracy estimate (Section 5, Theorem 5.1 below), that appears to be more general (in the convex case) than the one established in [38].

Figure 1: Theorem 1.4 and the non-evenness question on page 5 of arXiv:2411.17896.

3 Non-even local theorem

Let $\mathcal{H}^{2,\alpha}$ be the open subset of $C^{2,\alpha}(\mathbb{S}^{n-1})$ consisting of positive strict support functions, and define

$$\Xi_p : \mathcal{H}^{2,\alpha} \longrightarrow C^\alpha(\mathbb{S}^{n-1}), \quad \Xi_p(h) = h^{1-p} s_j(h, \cdot).$$

Theorem 1 (full local branch and C^0 -local uniqueness). *Fix $n \geq 3$, $1 \leq j \leq n-2$, $0 \leq p < 1$, and $0 < \alpha < 1$. Then the following hold.*

1. *There are neighborhoods U of 1 in $C^{2,\alpha}(\mathbb{S}^{n-1})$ and V of 1 in $C^\alpha(\mathbb{S}^{n-1})$ such that $\Xi_p : U \rightarrow V$ is a C^1 diffeomorphism. No parity condition is imposed.*
2. *After shrinking V , there is $\varepsilon > 0$ such that the branch solution $h_g \in U$ is the only classical support-function solution of (1) satisfying $\|h - 1\|_{C^0} < \varepsilon$.*
3. *If $g \in V$ is even, then h_g is even.*

These statements also hold at $(p, j) = (0, 1)$.

Proof. The derivative at the identity of the normalized j th elementary symmetric function on $(n-1) \times (n-1)$ matrices is $j/(n-1)$ times the trace. Consequently

$$D\Xi_p(1)[w] = \frac{j}{n-1} \Delta_{\mathbb{S}^{n-1}} w + (j+1-p)w. \quad (2)$$

On spherical harmonics of degree k , where $-\Delta_{\mathbb{S}^{n-1}}$ has eigenvalue $k(k+n-2)$, the multiplier in (2) is

$$\mu_k = j+1-p - \frac{j}{n-1} k(k+n-2). \quad (3)$$

We have $\mu_0 = j+1-p > 0$ and $\mu_1 = 1-p > 0$. For $k \geq 2$,

$$\mu_k \leq j+1-p - \frac{2jn}{n-1} < 0,$$

because $(n-1)(j+1) - 2jn = n-1 - j(n+1) \leq -2$. Thus (2) has no kernel. Schauder theory makes it a bounded isomorphism $C^{2,\alpha} \rightarrow C^\alpha$, and the Banach inverse function theorem proves part 1. Notice that the calculation includes $(p, j) = (0, 1)$.

We prove part 2. Fix $0 < \beta < \alpha$. If support functions h are uniformly C^0 -close to 1, their outer radii, hence their Lipschitz norms, are uniformly bounded. The elementary interpolation estimate

$$[h-1]_{C^\beta} \leq C \|h-1\|_{C^0}^{1-\beta} \text{Lip}(h-1)^\beta \quad (4)$$

shows that h is C^β -close to 1. If also g is C^α -close to 1, then

$$f := s_j(h, \cdot) = gh^{p-1} \quad (5)$$

is C^β -close to 1.

Consider now the classical area-measure map $\Psi(h) = s_j(h, \cdot)$. Its derivative at 1 is

$$D\Psi(1)[w] = \frac{j}{n-1} (\Delta_{\mathbb{S}^{n-1}} + n-1)w. \quad (6)$$

The kernel of (6) consists exactly of degree-one harmonics, which are translations $u \mapsto a \cdot u$. Restrict the domain to functions with zero degree-one component and the target to functions with zero first moment. (Every j th area measure has zero first moment.) Then (6) is an isomorphism. A second application of the inverse function theorem to (5) supplies a centered support function $\hat{h}$, $C^{2,\beta}$ -close to 1, with $s_j(\hat{h}) = f$.

Classical uniqueness for the j th area-measure problem says that two bodies with the same S_j differ by a translation [3]. Hence

$$h(u) = \hat{h}(u) + a \cdot u.$$

The degree-one component of a function C^0 -close to 1 is small, so $|a|$ is small. Choosing the C^0 and data neighborhoods sufficiently small places h in the inverse-function neighborhood obtained from part 1 with exponent β . The $C^{2,\alpha}$ branch solution also belongs to that neighborhood, so its $C^{2,\beta}$ uniqueness yields $h = h_g$.

Finally, if g is even then $u \mapsto h_g(-u)$ is another solution in U . Part 1 forces $h_g(-u) = h_g(u)$, proving part 3. $\square$

4 What remains global

Theorem 1 removes evenness from the entire near-ball branch, but it does not show that every solution for small data lies on that branch. The source obtains this global conclusion in the symmetric class by a C^0 non-collapsing estimate and compactness. Its upper-bound argument uses $h(u) \geq R|u \cdot v|$ and an origin-centered John ellipsoid, both consequences of central symmetry. Without symmetry one has only a one-sided support bound, and the negative-power density h^{p-1} can concentrate on the opposite hemisphere.

The centroid identity

$$\int_{\mathbb{S}^{n-1}} u g(u) h(u)^{p-1} du = 0$$

does determine the translation uniquely inside a fixed translation class: its derivative with respect to the translation vector is negative definite. It does not prevent degeneration of the centered shape. Adapting the constant-data maximum-principle proof of Chen [2] introduces first and second derivatives of g and likewise does not presently give the scale-free C^α compactness estimate. This is the precise remaining obstruction to the source's full non-even question.

5 Novelty and review status

A bounded search used the exact title and question, together with non-symmetric, near-isotropic, and L_p -Christoffel-Minkowski uniqueness terms. It found the source and the constant-data theorem, but no later answer or explicit statement of Theorem 1. Novelty confidence is modest: the full-space inverse-function step is short and may be regarded as implicit by specialists. The C^0 -local upgrade and the exact isolation of the global obstruction are the substantive additions.

Human-review focus. Check the normalization in (2); the zero-first-moment target for the $p = 1$ map; and the use of uniqueness of S_j up to translation. Do not read this packet as a global uniqueness theorem.

References

- [1] K. Patsalos and C. Saroglou, *A note on the L_p -Brunn-Minkowski inequality for intrinsic volumes and the L_p -Christoffel-Minkowski problem*, arXiv:2411.17896.
- [2] L. Chen, *Uniqueness of solutions to L_p -Christoffel-Minkowski problem for $p < 1$* , arXiv:1905.11043.
- [3] R. Schneider, *Convex Bodies: The Brunn-Minkowski Theory*, 2nd ed., Cambridge University Press, 2014.